Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MECHATRONICS

V SEMESTER B.TECH. MECHATRONICS

END SEMESTER EXAMINATION, NOV/DEC 2023

SUBJECT: THEORY OF MACHINES [MTE 3154]

(01/12/2023)

Time: 3 Hours **MAX. MARKS: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Data not provided may be suitable assumed.

Q. No		M	CO	PO	LO	BL
1A.	In a four link kinematic chain, link AB=8cm, BC=10cm, CD=12cm and DA=3cm. Does it satisfies Grashof's law?? Find all the inversions of this kinematic chain.	**2**	**1**	**2**	**2**	**L3**
1B.	Evaluate the three types of constrained motion with suitable examples	**3**	**1**	**1**	**1**	**L2**
1C.	Analyse the inversion obtained by fixing the slider of the following mechanism i) Single slider crank mechanism ii) Double slider crank mechanism. Explain each inversion with an application. Draw relevant sketches	**5**	**1**	**2**	**2**	**L4**
2A.	Sketch a Tchebicheff (Chebyshev) Mechanism. Mention the dimensions of links so that it can be used to trace a straight line	**2**	**1**	**1**	**1**	**L2**
2B	Design a four link mechanism to coordinate three positions of the input and the output links for the following angular displacements by inversion methods: - $\theta_{12} = 35^o$; $\theta_{13} = 80^o$; $\emptyset_{12} = 50^o$; $\emptyset_{13} = 80^o$	**3**	**5**	**3**	**3**	**L4**
2C	In a steam engine mechanism, stroke length is 30cm. Ratio of connecting rod length to crank radius is four. At a particular instant, angular velocity of crank is 20 rad/sec and is being retarded at the rate of 70 rad/sec^2. Angle turned by the crank from inner dead centre position is 30 degrees. Find out Acceleration of piston and angular acceleration of connecting rod. (Use Graphical Method, Write Graph sheet No in EPAD, Show all calculation in EPAD, and diagram in Graph sheet)	**5**	**2**	**4**	**3,4**	**L4**
3A	A flywheel absorbs 24kJ of energy on increasing its speed of 210 rpm, to 214 rpm. Determine its kinetic energy at 250 rpm	**2**	**3**	**1**	**1**	**L4**
3B	A flywheel which is rotating at a maximum speed of 250 rpm and is having a radius of gyration as 0.5m, is attached to a punching press. The punch is driven by a constant torque electric motor and punches 750 holes per hour. Each punching operation requires 14000 Nm of energy and takes 1.8 seconds. If the speed of the flywheel not to fall below 225 rpm find i) power of the motor, ii) mass of flywheel	**3**	**3**	**3**	**3**	**L4**

3C	A shaft running in bearings carries masses 20, 30 and 40kg, in planes A, B and C with centre of gravity from the axis of the shaft 30mm, 20mm and 15mm, respectively. The distances of planes B and C from A are 1000mm and 2000mm to the right of A. The relative angular position of the centre of gravity of the unbalanced masses are such that they are in static equilibrium. To obtain complete dynamic balance suitable masses are introduced in planes D and E with centre of gravity 100mm from the axis. D is 500mm to the left of A, and E 500mm to the right of C. Determine the position and magnitude of the balance masses (Use Graphical Method, Write Graph sheet No in EPAD, Show all calculation in EPAD, and diagram in Graph sheet)	**5**	**3**	**3**	**2,3**	**L4**
4A	Two equal gears are in mesh, having pressure angle of 14.5 degree. Addendum = 1 module. Find the minimum number of teeth required to avoid interference.	**2**	**4**	**2**	**2**	**L3**
4B	Show that Paucellier mechanism can be used to trace a straight line. Sketch the mechanism.	**4**	**5**	**2**	**2**	**L3**
4C	Two mating gears have 30 and 18 involute teeth of module 6mm and 20^0 pressure angle. The addendum on each wheel is one module. Pinion speed is 400 rpm. What is the sliding velocity when the pinion tooth tip touches the gear flank? Also find contact ratio.	**4**	**4**	**3**	**3**	**L4**
5A	Is it possible to build a mechanism with only three lower pairs? Justify your answer with a suitable sketch	**2**	**1**	**2**	**2**	**L4**
5B	Determine a suitable train of wheels to satisfy the requirements of a clock, the minute hand of which is fixed to a spindle and the hour hand to a sleeve rotating freely on the same spindle. The pitch is the same for all the wheels and each wheel has at least 14 teeth. The total number of teeth should be as small as possible.	**3**	**4**	**3**	**3**	**L4**
5C	In an Epicyclic gear train, the internal wheels C, D and the compound wheels A and B rotate independently about axis O. The wheels E and F rotate on pins fixed to the arm G. E gears with B and D and F gears with A and C. All wheels have the same module and number of teeth are $T_A = 52$; $T_B = 56$; $T_E=T_F=36$. a) Sketch the arrangement b) If arm G makes 200 rpm clockwise and D is fixed, find speed & direction of rotation of C c) If arm G makes 200 rpm clockwise and wheel D makes 20 rpm counter clockwise, find the speed of wheel C	**5**	**4**	**4**	**4**	**L4**

L1 – Remembering, L2 – Understanding, L3 - Applying, L4 – Analyzing, L5 – Evaluating, L6 - Creating